Yining **Li**

Research Scientist · LLM Post-training · Agentic Systems

Shanghai, China

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Profile

Research scientist at Shanghai AI Laboratory, leading a team focused on LLM post-training and AI-for-AI. Co-led SFT for InternLM 2/2.5/3 and led agentic coding post-training for Atria Dawn. Research spans training-data pipelines and agentic systems for automated model development and experimentation. Previously led MMPose development as a core contributor to OpenMMLab. Publications at NeurIPS, ICLR, ACL, and CVPR; 7,000+ Google Scholar citations. Ph.D., CUHK (MMLab).

Education

The Chinese University of Hong Kong

Hong Kong, China

Ph.D. in Information Engineering, MMLab — *Aug. 2015 - Sep. 2019*

- Advisors: Prof. Chen Change Loy and Prof. Xiaoou Tang

Tsinghua University

Beijing, China

B.S. in Automation — *Sep. 2010 - Jul. 2014*

Work Experience

Shanghai Artificial Intelligence Laboratory

Shanghai, China

Research Scientist — *Nov. 2021 - Present*

- Lead a research team focused on LLM post-training, training data, and autonomous AI R&D, setting research priorities and coordinating training, evaluation, and model releases.
- Co-led SFT for InternLM 2/2.5/3 and led agentic coding post-training for Atria Dawn, covering data pipelines and model fine-tuning.
- Previously led MMPose development and served as a core maintainer of OpenMMLab.

SenseTime

Shanghai, China

Research Scientist — *Sep. 2019 - Nov. 2021*

- Developed perception and scene-understanding models for autonomous driving, including risk prediction and human–vehicle interaction.
- Led two Honda-sponsored R&D projects, delivering three algorithmic solutions for hazard prediction, object-attribute perception, and human–vehicle interaction.

Projects

LLM Post-training for Coding and Alignment

2023 - Present

- **Atria Dawn post-training for coding.** Led post-training of Atria Dawn LLM on coding capabilities across software-engineering, terminal-use, and NL2Repo tasks; developed training-data pipelines and fine-tuned models built on GLM-5.2. [Project]
- **InternLM 2/2.5/3: supervised fine-tuning.** Co-led SFT for the InternLM series, focusing on instruction following, response quality, style consistency, and long-context reliability. [Project]
- **SFT data infrastructure.** Led development of LLM post-training data infrastructure spanning agent trajectory generation, SFT data curation, and high-quality response synthesis and evolution.

Autonomous AI-R&D Systems

2025 - 2026

- **TREX: automated LLM fine-tuning.** Led TREX, a multi-agent system that automates LLM fine-tuning through tree-based experimental search, from research planning and data preparation to training and evaluation. Introduced FT-Bench with 10 real-world fine-tuning tasks. Demonstrated competitive model-performance gains relative to published expert-designed pipelines on TOMG-Bench and OpenFinData. [Project]
- **Kernel-Smith: evolutionary kernel optimization.** Contributed to Kernel-Smith's evolutionary kernel-optimization framework, developing agents and training models for kernel optimization on NVIDIA and MetaX accelerators. [Paper] [Code]

Data Centric AI-for-AI

2024 - 2026

- **DataChef: RL-generated data recipes.** Developed DataChef to generate executable SFT data recipes and trained its 8B/32B models with GRPO. Designed a low-cost proxy reward that bypasses downstream training and correlates better with downstream performance than standard data-quality metrics on average (mean Pearson r = 0.59). Demonstrated substantial gains in recipe generation across 6 tasks, including coding, math, physics, etc. [Paper] [Code]
- **MIG: efficient instruction-data curation.** Developed MIG, an information-gain-based method for instruction-data selection that balances quality and diversity; achieved performance comparable to full-data SFT using 5% of Tulu 3 training data in the reported evaluation setting. [Project]
- **Auto Cherry-Picker: synthetic data for multimodal training.** Developed Auto Cherry-Picker, a synthetic-data framework combining language and diffusion models for downstream perception and reasoning tasks. [Project]

Tool-Use Agents: Benchmarks & Frameworks

2023 - 2025

- **GTA/GTA-2: general tool-agent benchmarks.** Designed GTA/GTA-2 benchmarks and evaluation pipelines for multimodal tool-use agents, spanning atomic tool use and open-ended workflows, with checkpoint-based evaluation of task completion. [Project]
- **AgentLego: multimodal tool API library.** Led development of AgentLego, a multimodal tool API library covering visual perception, image generation and editing, speech, and visual question answering, with integrations for Transformers Agents and Visual ChatGPT. [Code]

Open-Source Leadership

2021 - 2024

- **OpenMMLab core development and research leadership.** Served as a core maintainer of OpenMMLab, developing the data pipeline module for its 2.0 architecture. Led research on open-vocabulary perception within the OpenMMLab ecosystem. [Project]
- **MMPose: open-source pose estimation toolbox.** Led development of MMPose, a pose-estimation toolbox with 7.9k GitHub stars, supporting 30+ algorithms, 35+ datasets, modular pipelines, and cross-platform deployment. [Project]
- **RTMPose family: production-ready real-time pose estimation.** Developed and released a family of real-time models for 2D/3D body pose, crowded-scene pose, and whole-body keypoint estimation. Widely deployed in industry and used as a foundational pose extractor in AIGC systems (Animate Anyone, sd-webui-ControlNet, T2I Adapter). [Project]

Honors

2025 **Shanghai Oriental Talent Program,** Top Talent Project

2022 **Shanghai Rising-Star Program,** Category A (Academic Research)

Professional Activities

Academic Appointments

- Adjunct Supervisor, Shanghai Jiao Tong University (2025 - Present)
- Adjunct Supervisor, Shanghai Innovation Institute (2024 - 2025)

Service

- Conference Reviewer: CVPR, ECCV, ICLR, AAAI, NeurIPS, ACL, and COLM.
- Thesis Reviewer: Invited reviewer for doctoral and master's theses in China.

Invited Talks

- Introduction to Large Language Models — Shanghai Jiao Tong University, 2025.
- Introduction to InternLM — Shanghai Municipal Commission of Economy and Informatization, 2025.
- Data Efficiency in LLM Post-training — Shanghai Innovation Institute, 2024.
- Recent Research Highlights from OpenMMLab — SenseTime, 2024.
- Fundamentals of Computer Vision and Human Pose Estimation — Shanghai Jiao Tong University, 2023.

Publications

* Equal contribution † Corresponding author

Atria Dawn: The Dawn of Agentic Superintelligence — *arXiv 2026*

Atria Team — *Technical Report*

GTA-2: Benchmarking General Tool Agents from Atomic Tool-Use to Open-Ended Workflows — *IEEE TPAMI, Under Review*

J. Wang*, X. Liu*, **Y. Li**, S. Zhang, Y. Wang, Z. Shan, X. Le†, C. Chen†, X. Guan†, and D. Tao†

RouteMoA: Dynamic Routing without Pre-Inference Boosts Efficient Mixture-of-Agents — *ACL 2026*

J. Wang, H. Wu, Z. You, Y. Song, Y. Wang, Z. Shan, **Y. Li**, S. Zhang, X. Le, C. Chen†, X. Guan, and D. Tao — *Poster*

TREX: Automating LLM Fine-tuning via Agent-Driven Tree-based Exploration — *arXiv 2026*
Z. Ma*, G. Wang*, X. Xie*, Y. Chen, H. Du, B. Li, Y. Sun, W. Liu, K. Chen†, and **Y. Li**† — *Technical Report*

Kernel-Smith: A Unified Recipe for Evolutionary Kernel Optimization — *ICML 2026, DL4C Workshop*
H. Du*, Q. Ge*, J. Hu, A. Yang, Z. Cai, Z. Huang, S. Yuan, Q. Cheng, X. Xie, Y. Chen, **Y. Li**, J. Xie, H. Dong, Y. Wu, X. Huang, J. Yang, H. Wang, B. Zhou, B. Li†, Q. Guo†, and K. Chen† — *Poster*

DataChef: Cooking Up Optimal Data Recipes for LLM Adaptation via Reinforcement Learning — *Findings of EMNLP 2026*
Y. Chen, Z. Ma, X. Xie, **Y. Li**†, and K. Chen† — *Accepted*

MIG: Automatic Data Selection for Instruction Tuning by Maximizing Information Gain in Semantic Space — *Findings of ACL 2025*
Y. Chen, **Y. Li**†, K. Hu, Z. Ma, H. Ye, and K. Chen† — *Accepted*

Auto Cherry-Picker: Learning from High-quality Generative Data Driven by Language — *CVPR 2025*
Y. Chen, X. Li, **Y. Li**†, Y. Zeng, J. Wu, X. Zhao, and K. Chen† — *Poster*

MG-LLaVA: Towards Multi-Granularity Visual Instruction Tuning — *IEEE TCSVT, 2025*
X. Zhao, X. Li, H. Duan, H. Huang, **Y. Li**, K. Chen†, and H. Yang† — *Journal Article*

RMP-SAM: Towards Real-Time Multi-Purpose Segment Anything — *ICLR 2025*
S. Xu*, H. Yuan*, Q. Shi, L. Qi, J. Wang, Y. Yang, **Y. Li**, K. Chen, Y. Tong, B. Ghanem, X. Li†, and M.-H. Yang — *Oral*

MotionBooth: Motion-Aware Customized Text-to-Video Generation — *NeurIPS 2024*
J. Wu, X. Li, Y. Zeng, J. Zhang, Q. Zhou, **Y. Li**, Y. Tong, and K. Chen — *Spotlight*

Efficient LLM Jailbreak via Adaptive Dense-to-sparse Constrained Optimization — *NeurIPS 2024*
K. Hu*, W. Yu*, **Y. Li**, K. Chen, T. Yao, X. Li, W. Liu, L. Yu, Z. Shen, and M. Fredrikson — *Poster*

InternLM-XComposer2-4KHD: A Pioneering Large Vision-Language Model Handling Resolutions from 336 Pixels to 4K HD — *NeurIPS 2024*
X. Dong*, P. Zhang*, Y. Zang*, Y. Cao, B. Wang, L. Ouyang, S. Zhang, H. Duan, W. Zhang, **Y. Li**, H. Yan, Y. Gao, Z. Chen, X. Zhang, W. Li, J. Li, W. Wang, K. Chen, C. He, X. Zhang, J. Dai, Y. Qiao, D. Lin, and J. Wang† — *Poster*

InternLM2 Technical Report — *arXiv 2024*
InternLM Team — *Technical Report*

InternLM-XComposer2: Mastering Free-Form Text-Image Composition and Comprehension in Vision-Language Large Model — *arXiv 2024*
X. Dong*, P. Zhang*, Y. Zang*, Y. Cao, B. Wang, L. Ouyang, X. Wei, S. Zhang, H. Duan, M. Cao, W. Zhang, **Y. Li**, H. Yan, Y. Gao, X. Zhang, W. Li, J. Li, K. Chen, C. He, X. Zhang, Y. Qiao, D. Lin, and J. Wang† — *Technical Report*

GTA: A Benchmark for General Tool Agents — *NeurIPS D&B 2024*
J. Wang, Z. Ma, **Y. Li**, S. Zhang, C. Chen†, K. Chen†, and X. Le† — *Poster*

RTMW: Real-Time Multi-Person 2D and 3D Whole-body Pose Estimation — *arXiv 2024*
T. Jiang*, X. Xie*, and **Y. Li** — *Technical Report*

Open-Vocabulary SAM: Segment and Recognize Twenty-thousand Classes Interactively — *ECCV 2024*
H. Yuan, X. Li, C. Zhou, **Y. Li**, K. Chen, and C. C. Loy — *Poster*

RTMO: Towards High-Performance One-Stage Real-Time Multi-Person Pose Estimation — *CVPR 2024*
P. Lu, T. Jiang, **Y. Li**, X. Li, K. Chen†, and W. Yang† — *Poster*

OMG-Seg: Is One Model Good Enough for All Segmentation? — *CVPR 2024*
X. Li†, H. Yuan, W. Li, H. Ding, S. Wu, W. Zhang, **Y. Li**, K. Chen, and C. C. Loy — *Poster*

Towards Language-Driven Video Inpainting via Multimodal Large Language Models — *CVPR 2024*
J. Wu, X. Li†, C. Si, S. Zhou, J. Yang, J. Zhang, **Y. Li**, K. Chen, Y. Tong, Z. Liu, and C. C. Loy — *Poster*

InternLM-XComposer-2.5: A Versatile Large Vision Language Model Supporting Long-Contextual Input and Output — *arXiv 2024*

P. Zhang*, X. Dong*, Y. Zang*, Y. Cao, R. Qian, L. Chen, Q. Guo, H. Duan, B. Wang, L. Ouyang, S. Zhang, W. Zhang, **Y. Li**, Y. Gao, P. Sun, X. Zhang, W. Li, J. Li, W. Wang, H. Yan, C. He, X. Zhang, K. Chen, J. Dai, Y. Qiao, D. Lin, and J. Wang† — *Technical Report*

MMBench-Video: A Long-Form Multi-Shot Benchmark for Holistic Video Understanding — *NeurIPS D&B 2024*

X. Fang*, K. Mao*, H. Duan*†, X. Zhao, **Y. Li**, D. Lin, and K. Chen† — *Poster*

An Open and Comprehensive Pipeline for Unified Object Grounding and Detection — *arXiv 2024*

X. Zhao, Y. Chen, S. Xu, X. Li, X. Wang, **Y. Li**, and H. Huang — *Technical Report*

RTMPose: Real-Time Multi-Person Pose Estimation Based on MMPose — *arXiv 2023*

T. Jiang, P. Lu, L. Zhang, N. Ma, R. Han, C. Lyu, **Y. Li**, and K. Chen — *Technical Report*

Deep Imbalanced Learning for Face Recognition and Attribute Prediction — *IEEE TPAMI, 2019*

C. Huang, **Y. Li**, C. C. Loy, and X. Tang

Dense Intrinsic Appearance Flow for Human Pose Transfer — *CVPR 2019*

Y. Li, C. Huang, and C. C. Loy — *Poster*

Learning to Disambiguate by Asking Discriminative Questions — *ICCV 2017*

Y. Li, C. Huang, X. Tang, and C. C. Loy — *Poster*

Learning Deep Representation for Imbalanced Classification — *CVPR 2016*

C. Huang, **Y. Li**, C. C. Loy, and X. Tang — *Spotlight*

Human Attribute Recognition by Deep Hierarchical Contexts — *ECCV 2016*

Y. Li, C. Huang, C. C. Loy, and X. Tang — *Poster*